

Programme	B. Sc. Mathematics Honours			
Course Code	MAT7CJ404			
Course Title	LINEAR ALGEBRA			
Type of Course	Major			
Semester	VII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial per week	Practicum per week	Total Hours
	4	3	2	75
Pre-requisites	1. Mathematical Logic and necessary exposure to set theory. 2. Matrices and Determinants 3. Systems of Linear Equations and their solutions			
Course Summary	Vector spaces in the abstract are introduced. Linear transformations are introduced as structure preserving maps between them. Representation of linear transformations as matrices is discussed. The algebraic dual and double dual space of a vector space are studied in some detail. The concept of the transpose of a linear transformation is introduced and discussed as well. The course then passes on to spectral theory on finite dimensional spaces, introducing characteristic values and vectors. After an extended discussion leading up to the characterisation of diagonalisable and triangulable operators, an elementary decomposition of a linear operator is established. The course ends with a short discussion of inner products and inner product spaces.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Analyse and apply the concepts of vector spaces, subspaces, and bases to solve problems involving linear independence and dimensionality.	An	P	Internal Exam/Assignment/Seminar/ Viva/Report/ End Sem Exam
CO2	Evaluate the properties of linear transformations and their algebraic representations using matrices.	E	C	Internal Exam/Assignment/Seminar/ Viva/Report/ End Sem Exam
CO3	Synthesize the concepts of linear functionals, the double dual space, and the transpose of linear transformations to understand advanced topics in linear algebra and apply them to canonical forms	E	P	Internal Exam/Assignment/Seminar/ Viva/Report/ End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook		Linear Algebra, Kenneth Hoffman and Ray Kunze, 2 nd Edition, Prentice Hall of India, 1991.		
Module	Unit	Content	Hrs (45+30)	External Marks (70)
I	Vector Spaces		12	Min.15
	1	Section 2.1 – Vector Spaces		
	2	Section 2.2 – Subspaces		
	3	Section 2.3 – Bases and Dimension – up to Theorem 5		
	4	Section 2.3 – Bases and Dimension – rest of the section starting from Theorem 5		
	5	Section 2.4 – Coordinates – up to and including Theorem 7		
	6	Section 2.4 – Coordinates – rest of the section		
II	Linear Transformations		11	Min.15
	7	Section 3.1 – Linear Transformations – upto and including Example 7		
	8	Section 3.1 – Linear Transformations – rest of the section.		
	9	Section 3.2 – The Algebra of Linear Transformations – up to and including Theorem 5		
	10	Section 3.2 – The Algebra of Linear Transformations – rest of the section		
	11	Section 3.3 – Isomorphism		
	12	Section 3.4 – Representation of Transformations by Matrices – up to and including Example 15		
III	Linear Transformations		11	Min.15
	13	Section 3.4 – Representation of Transformations by Matrices – rest of the section		
	14	Section 3.5 – Linear Functionals – upto and including Example 22.		
	15	Section 3.5 – Linear Functionals – rest of the section.		
	16	Section 3.6 – The Double Dual – upto and including Theorem 18.		
	17	Section 3.6 – The Double Dual – the rest of the section		
	18	Section 3.7 – The Transpose of a Linear Transformation – up to and including Theorem 22		
	19	Section 3.7 – The Transpose of a Linear Transformation – rest of the section.		
IV	Elementary Canonical Forms		11	Min.15
	20	Section 6.1 and 6.2 – Introduction and Characteristic Values		
	21	Section 6.3 – Annihilating Polynomials (Proof of Theorem 4 omitted)		
	22	Section 6.4 – Invariant Subspaces.		

V	Practicum		30	-
	The goal is for the students to learn the following selected topics in 10 practicum sessions of three hours each via self-study and group activities. The lecturer may assist by running group discussions, supervising class seminars and referring library books for self-study and note preparations.			
	1	Section 1.3 – Matrices and Elementary Row Operations		
	2	Section 1.4 – Row Reduced Echelon Matrices		
	3	Section 1.5 – Matrix Multiplication		
	4	Section 1.6 – Invertible Matrices		
	5	Section 6.4 – Triangulation and Diagonalisation		
	6	Section 6.6 – Direct-sum Decompositions		
	7	Section 6.7 – Invariant Direct Sums		
	8	Section 8.1 – Inner Products		
	9	Section 8.2 – Inner Product Spaces		
	10	Section 6.8 – The Primary Decomposition Theorem		
References 1. Finite Dimensional Vector Spaces, P. R. Halmos, Narosa Pub House, 1980.. 2. Linear Algebra, S. Lang, Addison Wesley Pub Company, 1972. 3. Topics in Algebra, I. N. Herstein, John Wiley & Sons, 2006. 4. Linear Algebra, R. R. Stoll & E. T. Wong, Academic Press International Edition, 1968.				
Suggested Programming Exercises for Practicum : 1. Form a four-dimensional vector space over $\mathbb{Q}$. Take two vectors from this, find its span. (Chapter VS, Ref (1)) 2. Find basis of the vector subspace found in the above question. (Chapter VS, Ref (1)) 3. Take some elements from this vector space, test for linear independence. (Chapter V Section LI, Ref (1)) 4. Form two vector spaces over $\mathbb{Q}$. Define symbolic linear transformations between them, find the image of selected elements under it. (Chapter LT, Ref (1)) 5. Define linear transformations (LT) from matrices. (Chapter LT, Ref (1)) 6. Check if linear transformation is injective (Section ILT , Ref (1)) 7. Define two LT, add them. Find the individual matrices of these with respect to certain bases. Verify that the matrix of the sum of LT is the sum of matrices of individual LT .(Section OLT, , Ref (1)) 8. Find the kernel of an LT, find its nullity. (Section ILT, Ref (1)) 9. Find inverse of LT (Section IVLT, Ref (1)) 10. Take a matrix, find Eigenvalues, Eigen vectors, check if it is				

diagonalizable, diagonalize if it is. (Chapter E ILT, Ref (1))

References

1. Robert A. Beezer, Sage for Linear Algebra A Supplement to A First Course in Linear Algebra <http://linear.ups.edu/sage-fcla.html>
2. Sang-Gu Lee *et al.*, Linear Algebra with Sage https://www.researchgate.net/publication/280093747_Linear_Algebra_with_Sage_BigBook_Free_e-book_English_Version_All

***70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	2	1	3	0	3	1	3	0	2
CO 2	3	3	2	1	3	0	3	2	3	0	2
CO 3	3	3	2	2	3	0	3	2	3	0	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Report
- Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment/ Report	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓